

EXPERIMENT 17

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Aim

To write a menu-driven C program using **switch-case** to perform digit-based operations on a positive integer. A separate **void user-defined function** is used for each menu option, and each function reads its own input, processes it, and displays the result.

Required Operations

1. Generate the largest possible number using the digits.
2. Generate the smallest possible number using the digits.
3. Swap the first and last digits.
4. Check whether the number is a palindrome.

Algorithm

Main Algorithm

1. Start.
2. Display the four menu options.
3. Read the user's choice.
4. Use switch-case to call one function: largestNumber(), smallestNumber(), swapFirstLast(), or checkPalindrome().
5. Each function reads a positive integer and performs its specified operation.
6. For any choice other than 1-4, display "Invalid choice".
7. Stop.

Option 1 - Largest possible number

Count the occurrences of digits 0-9. Print digits from 9 down to 0 according to their frequencies.

Option 2 - Smallest possible number

Count digit occurrences. The first digit cannot be zero, so print the smallest available non-zero digit first, then all zeros, followed by the remaining digits in ascending order.

Option 3 - Swap first and last digits

Read the number as a string. Exchange the first and last characters, then display the resulting number. Leading zeros produced by the swap are retained in the displayed digit arrangement.

Option 4 - Palindrome check

Reverse the number mathematically and compare the reversed value with the original. If both are equal, it is a palindrome.

Test Case Table

Test	Choice	Input	Expected Output
1	1	31042	Largest possible number = 43210
2	1	99801	Largest possible number = 99810
3	2	31042	Smallest possible number = 10234
4	2	90021	Smallest possible number = 10029

Test	Choice	Input	Expected Output
5	3	12345	After swapping = 52341
6	3	1020	After swapping = 0021
7	4	12321	Palindrome number
8	4	12345	Not a palindrome number
9	5	-	Invalid choice

C Program

```
#include <stdio.h>
#include <string.h>

void largestNumber(void);
void smallestNumber(void);
void swapFirstLast(void);
void checkPalindrome(void);

int main(void)
{
    int choice;

    printf("1. Generate largest possible number\n");
    printf("2. Generate smallest possible number\n");
    printf("3. Swap first and last digits\n");
    printf("4. Check palindrome number\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1: largestNumber(); break;
        case 2: smallestNumber(); break;
        case 3: swapFirstLast(); break;
        case 4: checkPalindrome(); break;
        default: printf("Invalid choice\n");
    }
    return 0;
}

void largestNumber(void)
{
    char n[50];
    int count[10] = {0}, i, d;

    printf("Enter a positive integer: ");
    scanf("%49s", n);

    if (n[0] == '-' || strcmp(n, "0") == 0) {
        printf("Please enter a positive integer.\n");
        return;
    }

    for (i = 0; n[i] != '\0'; i++)
        count[n[i] - '0']++;

    printf("Largest possible number = ");
    for (d = 9; d >= 0; d--)
        while (count[d]-- > 0)
            printf("%d", d);
    printf("\n");
}

void smallestNumber(void)
{
    char n[50];
    int count[10] = {0}, i, d, first = -1;

    printf("Enter a positive integer: ");
    scanf("%49s", n);

    if (n[0] == '-' || strcmp(n, "0") == 0) {
        printf("Please enter a positive integer.\n");
        return;
    }

    for (i = 0; n[i] != '\0'; i++)
        count[n[i] - '0']++;

    for (d = 1; d <= 9; d++)
        if (count[d] > 0) {
            first = d;
            break;
        }

    printf("Smallest possible number = ");
    printf("%d", first);
    count[first]--;

    while (count[0]-- > 0)
```

```

        printf("0");

    for (d = 1; d <= 9; d++)
        while (count[d]-- > 0)
            printf("%d", d);

    printf("\n");
}

void swapFirstLast(void)
{
    char n[50], temp;
    int len;

    printf("Enter a positive integer: ");
    scanf("%49s", n);

    if (n[0] == '-' || strcmp(n, "0") == 0) {
        printf("Please enter a positive integer.\n");
        return;
    }

    len = strlen(n);
    if (len > 1) {
        temp = n[0];
        n[0] = n[len - 1];
        n[len - 1] = temp;
    }

    printf("After swapping = %s\n", n);
}

void checkPalindrome(void)
{
    long long n, original, reverse = 0, digit;

    printf("Enter a positive integer: ");
    scanf("%lld", &n);

    if (n <= 0) {
        printf("Please enter a positive integer.\n");
        return;
    }

    original = n;
    while (n > 0) {
        digit = n % 10;
        reverse = reverse * 10 + digit;
        n /= 10;
    }

    if (original == reverse)
        printf("Palindrome number\n");
    else
        printf("Not a palindrome number\n");
}

```

Compilation and Execution

Save the program as: experiment17.c

Compile: gcc experiment17.c -o experiment17

Execute: ./experiment17

Sample run - largest possible number:

```

Enter your choice: 1
Enter a positive integer: 31042
Largest possible number = 43210

```

Sample run - smallest possible number:

```

Enter your choice: 2
Enter a positive integer: 31042
Smallest possible number = 10234

```

Sample run - palindrome:

```

Enter your choice: 4
Enter a positive integer: 12321

```

Result

The menu-driven C program was prepared using switch-case and four separate void user-defined functions. The program generates the largest and smallest possible numbers from the given digits, swaps the first and last digits, and checks whether the entered positive integer is a palindrome.